GOOGLE DATA ANALYTICS CAPSTONE CASE STUDY

Converting Casual Riders to Annual Members

Data-driven marketing strategies based on 400K+ trip records to drive long-term business growth.

Slide created by: Joachim Armookyi Kwofie

Prepared for Lily Moreno & Executive Team

OVERVIEW

Executive Summary

Strategic Goal

Convert non-subscription casual riders into high-margin annual members.

Key Discovery

Members ride for weekday commutes; casuals ride for weekend leisure.

Core Recommendation

Target casuals with targeted weekend promotions and digital push campaigns.

CONTEXT

The Business Challenge

Maximizing Annual Memberships

Annual members generate more consistent, predictable revenue than single-pass riders.

Target Audience Transition

Casual riders are already familiar with Cyclistic; marketing directly to them is cost-effective.

Data-Driven Action

Identify distinct usage behaviors to tailor marketing messages effectively.

> **"Our future growth relies on converting casual riders into long-term annual subscribers."**
>
> — Lily Moreno, Director of Marketing

METHODOLOGY

Data Processing Pipeline

STEP 1

Data Ingestion

420,000+ raw ride logs imported into Google Sheets.

STEP 2

Metric Creation

Calculated ride length in minutes and extracted day of week.

STEP 3

Data Scrubbing

Removed false system trips (< 1 min).

STEP 4

Outlier Capping

Filtered extreme outliers (> 1,440 mins) for 418,919 clean records.

ANALYSIS FINDING 01

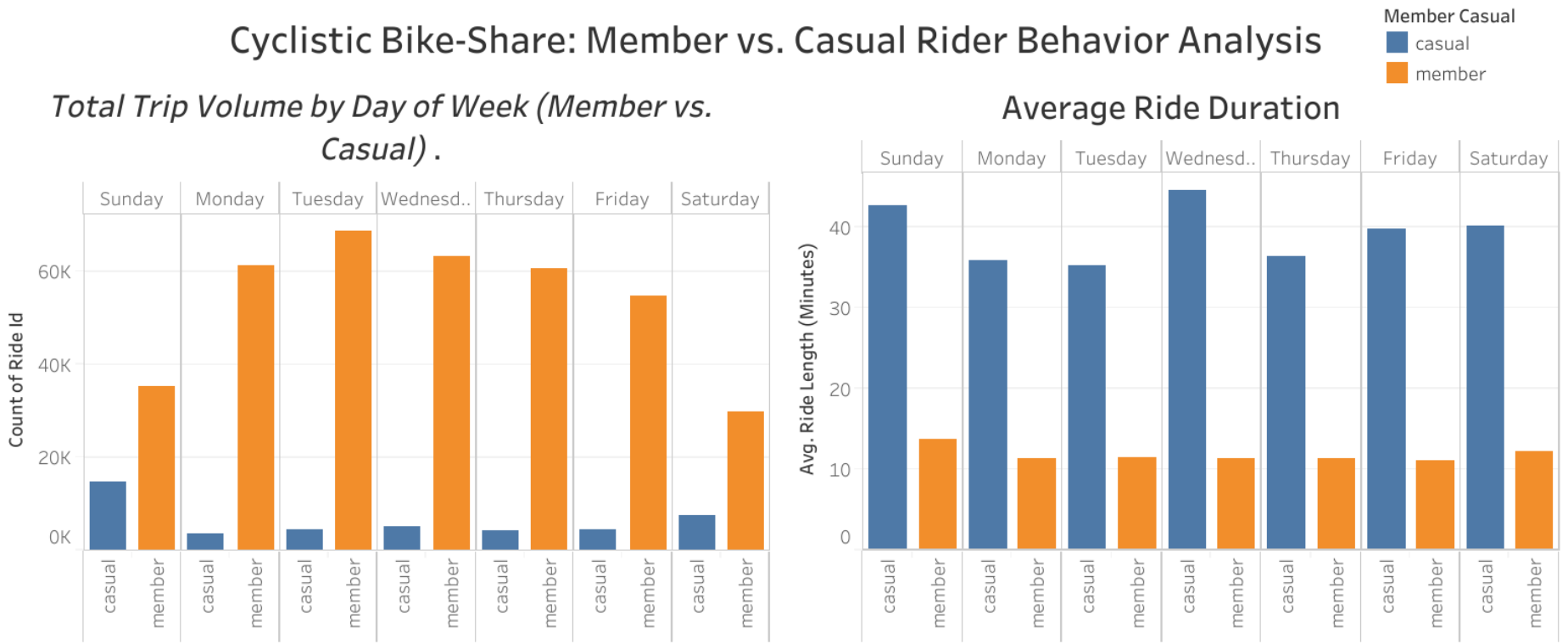
Trip Volume & Frequency Patterns

80%+

Member Peak Volume (Tues–Wed)

2.5x

Casual Weekend Surge Factor



Key Takeaway: Members use bikes for routine daily commuting, while casual users heavily concentrate activity on weekends.

ANALYSIS FINDING 02

Average Ride Duration Disparity

CASUAL RIDERS

35–44 min

Average trip length across all days

Behavior: Leisure rides, sightseeing, and flexible exploration.

ANNUAL MEMBERS

11–13 min

Average trip length across all days

Behavior: Efficient, point-to-point daily transit commuting.

DATA INSIGHTS

Weekly Trip Distribution Summary



ACTION PLAN

Strategic Recommendations

01

Weekend Pass Upgrade

Credit casual weekend trip costs directly toward annual subscription sign-ups.

02

Seasonal Pass Option

Introduce a Summer Member Pass targeting long-duration recreational riders.

03

Targeted In-App Prompts

Trigger conversion notifications for riders exceeding 30+ minute trips.

EXECUTION

Implementation & Next Steps

1. **A/B Testing:** Test conversion messaging on mobile app users following weekend rides.
2. **Digital Campaign:** Launch social media ad push highlighting long-ride cost savings.
3. **Station Marketing:** Place promotional signs at popular weekend recreational stations.

Questions & Discussion

Thank you for your time. Let's discuss how we can put these insights into action.